# How to Create a Google Cloud Project and Generate Your Gemini API Key

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## Before you start

You will create your own Google Cloud project and a Gemini API key inside it, all from Google AI Studio, in about 5 minutes. You need this key to run every Gemini notebook in the course.

**Two terms to know**

- **Google Cloud project**: a container that holds your API keys, usage limits (quota) and billing settings. Every Gemini API key belongs to exactly one project.
- **API key**: a long secret code (it starts with `AIza`) that tells Google the request is coming from your project. Treat it like a password.

**What you need**

- A personal Google (Gmail) account. Work or school accounts often block key creation; use a personal one if you hit that.
- A desktop or laptop browser (Chrome recommended).
- No credit card. The free tier is enough for this course.

## Step 1: Sign in to Google AI Studio

1. Open aistudio.google.com in your browser.
2. Click **Sign in** and log in with your personal Google account.
3. If a Terms of Service pop-up appears, tick the box to accept the terms and click **Continue**. The second checkbox (marketing emails) is optional.
4. You now see the AI Studio home screen with a left-side panel.

**Note for first-time users:** after you accept the terms, AI Studio may automatically create a default project and a default key for you. That key works, but for this course we create our own named project so you always know which key belongs to which work.

## Step 2: Create your project

1. In the left panel, click **Dashboard**, then click **Projects**.
2. Click the button to create a new project (labelled **Create project** or **New project**).

3. Type a clear project name, for example `DDS-Gemini-Course`. Use letters, numbers and hyphens only.
4. Click **Create**. Wait a few seconds until the new project appears in the Projects list.

**Alternative (if you do not see a create button):** go to Step 3 directly. When you click **Create API key**, the project drop-down in that pop-up offers an option to create a new project. Choose it and give it the same name.

**Already have a Google Cloud account?** AI Studio will not create a default project for you. Either create a new one as above, or bring in an existing one: **Dashboard → Projects → Import projects**, search for your project, select it and click **Import**.

## Step 3: Generate your API key

1. In the left panel, open **Dashboard → API Keys** (you can also go straight to aistudio.google.com/app/apikey).
2. Click **Create API key** (top right).
3. If asked for a key name, type something descriptive, for example `dds-course-key`.
4. In the project drop-down, select the project you created in Step 2 (`DDS-Gemini-Course`).
5. Click **Create key**. The key appears within a few seconds.
6. Click the **Copy** icon next to the key and paste it somewhere private straight away (see Step 4).

**Check:** a valid key starts with `AIza` followed by a long string of letters and numbers.

One key works for all Gemini models in your project. You do not need a separate key per model; your code chooses the model. New keys made in AI Studio are restricted to the Gemini API by default, which is the safe setting.

## Step 4: Save the key in Google Colab and test it

Store the key in Colab Secrets so it never appears in your notebook code.

1. Open your Colab notebook and click the **key icon** (Secrets) in the left sidebar.
2. Click **Add new secret**.
3. Name: `GEMINI_API_KEY`. Value: paste your key.
4. Switch on **Notebook access** for that secret.
5. Run the test cell below. If Gemini replies with a sentence, you are ready.

```
!pip install -q -U google-genai

from google.colab import userdata
from google import genai

client = genai.Client(api_key=userdata.get("GEMINI_API_KEY"))

response = client.models.generate_content(
    model="gemini-flash-latest",
    contents="Explain what an API key is in one sentence."
)
print(response.text)
```

**Working on your own laptop instead of Colab?** Set an environment variable named `GEMINI_API_KEY`; the Gemini libraries pick it up automatically.

- macOS / Linux (add to `~/.bashrc` or `~/.zshrc`): `export GEMINI_API_KEY="your_key_here"`
- Windows (Command Prompt): `setx GEMINI_API_KEY "your_key_here"`, then open a new terminal.

Model names change over time. If the model in the test cell is not found, open AI Studio, copy a current Flash model name from the model picker and paste it into `model=`.

## Keep your key safe

Anyone with your key can use your project's quota and, if billing is on, run up charges.

- Never paste the key directly into notebook code, slides, screenshots or WhatsApp groups.
- Never push it to GitHub. Use Colab Secrets or environment variables instead.
- If a key leaks, delete it on the **API Keys** page and create a new one.
- Do not enable billing for this course unless your instructor asks you to.

## Troubleshooting

| Problem | What to do |
| --- | --- |
| **Create API key** button is greyed out, or "You do not have permission to create a key in this project" | You are likely on a work or school account. Sign in with a personal Gmail account, or create a new project that is not linked to an organisation. |

| Problem | What to do |
| --- | --- |
| Your project does not appear in the drop-down | Go to **Dashboard → Projects → Import projects**, select it and click **Import**, then return to **API Keys**. |
| "API key not valid" error | Re-copy the key with no extra spaces, and check the secret name is exactly `GEMINI_API_KEY`. |
| Colab says the secret is not found | Open the Secrets panel and switch on **Notebook access** for `GEMINI_API_KEY`. |
| "Model not found" error | Replace the model name with a current one from the AI Studio model picker. |
| "Quota exceeded" or 429 error | You hit the free-tier rate limit. Wait a minute and run the cell again. |

## Checklist before the session

- [ ] Signed in to Google AI Studio with a personal Google account
- [ ] Created a project named for this course
- [ ] Generated an API key in that project (starts with `AIza`)
- [ ] Saved the key as `GEMINI_API_KEY` in Colab Secrets with Notebook access on
- [ ] Ran the test cell and got a reply from Gemini

## Sources

- Using Gemini API keys – Google AI for Developers
- Google AI Studio